

FOOD ACCESS & NUTRITION IN THE UNITED STATES

Your *zip code* determines your diet — not your willpower

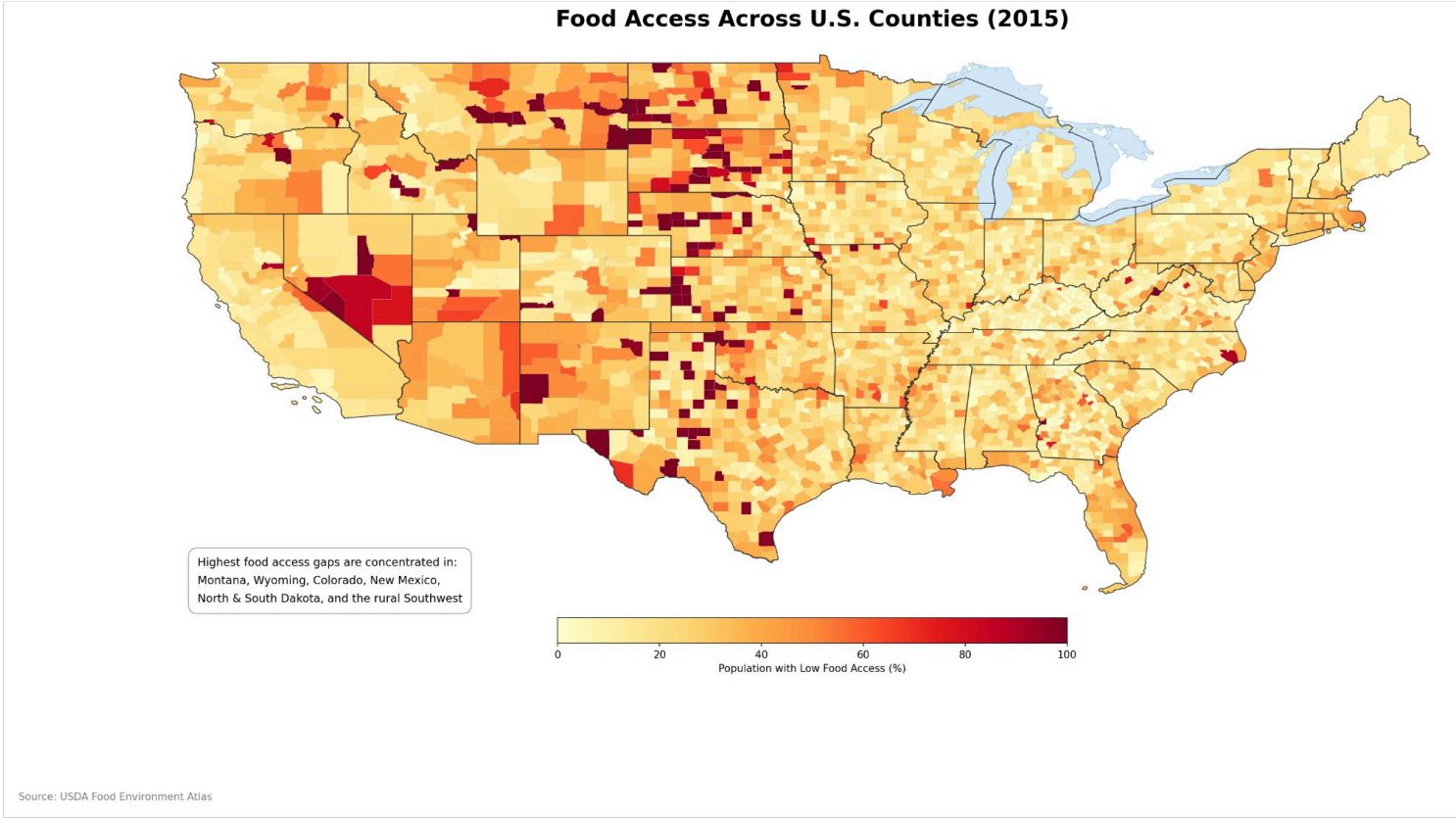
Where you live and what you earn shape your access to healthy food long before personal choice enters the picture. This project uses USDA county-level data to make that invisible gap visible.

3,119 U.S. counties analyzed	10× gap between best and worst food access states (DC vs. ND)	+3.1 pts higher adult obesity in lowest vs. highest income counties	780 counties in worst access quartile — worst on every health metric
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FINDING 1 — GEOGRAPHY OF ACCESS

Food deserts are not scattered randomly — they follow the map of rural poverty

The darkest counties — where 60–100% of residents lack adequate grocery access — cluster in the Great Plains, Mountain West, and rural Southwest. The coasts and urban corridors stay light. This is a structural problem, not a personal one.



FINDING 2 — THE HEALTH TOLL

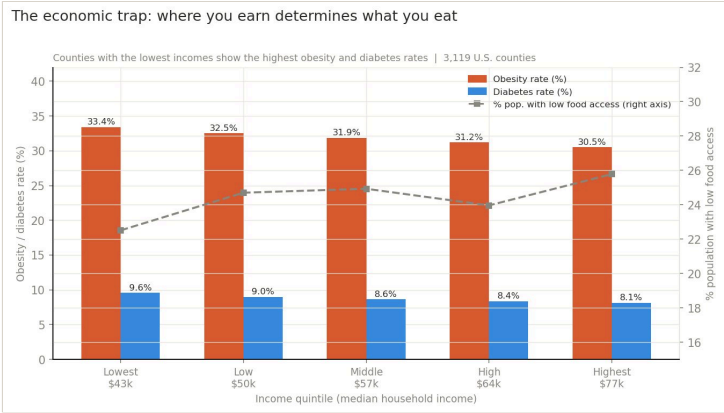
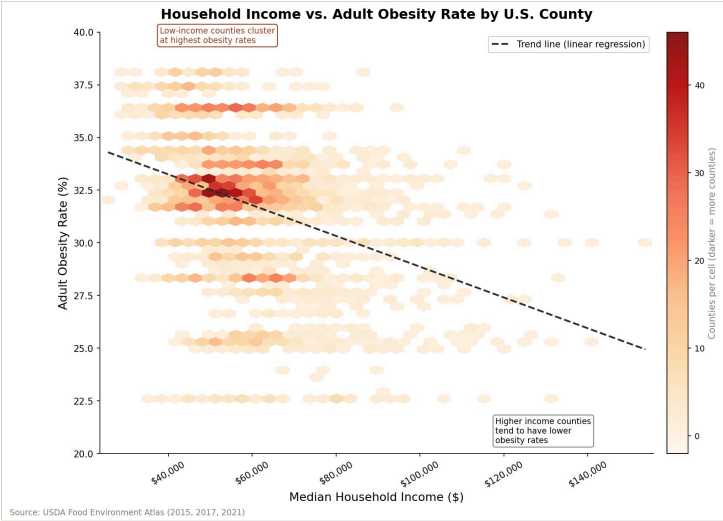
As income falls, obesity climbs — across every county in America

3,119 counties. The trend line says it all: lower income predicts higher obesity. The densest cluster sits at low income, high obesity — not because poor people make bad choices, but because their options are constrained by circumstance.

FINDING 3 — THE ECONOMIC TRAP

Every step down the income ladder means worse health outcomes

From the highest to the lowest income quintile, obesity rises from 30.5% to 33.4% and diabetes from 8.1% to 9.6%. The gradient is consistent and monotonic — income is doing real work here, not coincidence.

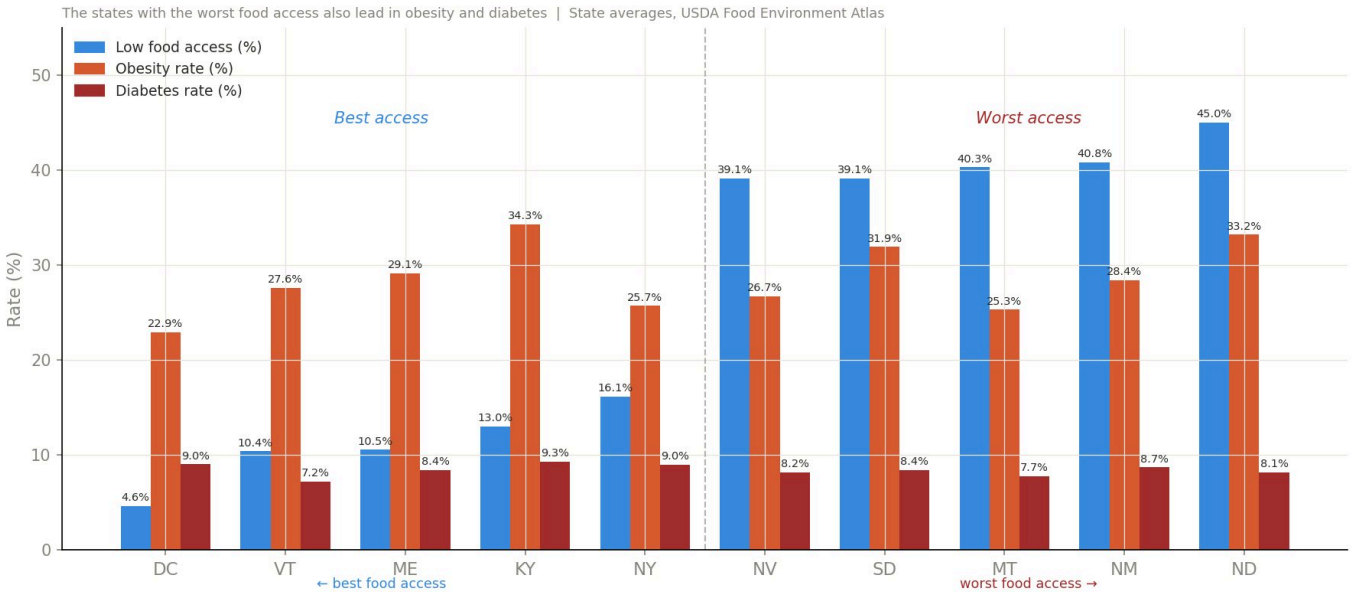


FINDING 4 — ZIP CODE AS DESTINY

The states with the worst food access also lead in obesity and diabetes

DC, Vermont, and Maine have the best grocery access in the country — and comparatively lower obesity rates. North Dakota, New Mexico, and Montana sit at the opposite extreme on both measures. The pattern is not subtle.

Your zip code determines your diet — and your health



Source: USDA Food Environment Atlas (2017, 2019)

The data is clear

Every visualization in this project points the same direction. The counties with the least food access are also the poorest, the sickest, and the most geographically isolated. That is not coincidence.

Methods

USDA Food Environment Atlas merged by county FIPS code. Sentinel values removed. Income quintiles via `pd.qcut()`. Visualizations in Python — `matplotlib`, `geopandas`, `seaborn`. County shapefiles from U.S. Census TIGER/Line 2019.

Limitations

Correlation, not causation. County-level aggregates mask within-county variation. Food access measured by distance to store, not affordability. Health and economic data span different years (2015–2021).